N Quiz

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Portland State UNIVERSITY

Codefest 8

Hardware for Artificial Intelligence and Machine Learning (HW4AI)

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Overview

This codefest has a single CMAN challenge. No CLLM this week because M3 is due the same Sunday and your time is better spent on synthesis. CMAN must be completed without any AI assistance.

- **Topic**: Address-Event Representation (AER) — bandwidth analysis for event-driven neuromorphic communication.
- **Deliverable** is committed to your GitHub repository by Sunday May 24, 11:59 pm.

CMAN — AER bandwidth analysis

You are designing the off-chip output interface for a spiking neural network (SNN) accelerator. The SNN has N = 1024 output neurons, each with a mean firing rate of f = 50 Hz (spikes per second). Spikes are communicated to the host using Address-Event Representation (AER): when a neuron fires, the chip emits a packet containing the neuron address plus framing overhead. There is no clock-driven frame; packets are emitted asynchronously as spikes occur. Assume firing is independent across neurons (Poisson-like statistics).

Tasks

1. Compute the mean aggregate spike rate R (spikes/second) for the whole output layer. Show the formula and substitute values: R = N × f. = (1024)(50) = 51200 spikes/sec
2. Each AER packet contains a 10-bit neuron address (since $\log_2(1024) = 10$), a 6-bit timestamp, and a 4-bit framing/parity overhead, for 20 bits per packet total. Compute the mean required AER bandwidth B in bits/second and convert to Mbit/s. Show the formula: B = R × 20. = (51200)(20) = 1,02M bit/s
3. Compare your B to each of the standard interfaces from the M1 table: SPI (≤50 Mbit/s), I²C (≤3.4 Mbit/s), AXI4-Lite (assume 100 Mbit/s effective for a narrow bus). For each interface, state whether it can sustain the mean rate, and identify which is the lowest-complexity interface that suffices.
4. Bursts matter. Suppose stimulus arrives that causes 25% of the 1024 neurons to fire within a 1 ms window (synchronous burst). Compute the peak instantaneous bandwidth required in Mbit/s during that window. Compare to the mean bandwidth from task 2 and identify the burst-to-mean ratio. State whether the interface chosen in task 3 can absorb the burst, or whether buffering is required (and roughly how deep).
5. Frame-based comparison. A conventional (non-AER) readout would sample all 1024 neurons every 1 ms regardless of activity, sending 1 bit per neuron per sample. Compute the frame-based bandwidth in Mbit/s and the AER-to-frame bandwidth ratio at the mean firing rate. State the firing rate f_crossover at which AER and frame-based bandwidths are equal (set them equal and solve for f). Briefly state in one sentence what this implies for when AER is the right choice.

$B = \frac{1024000}{1000000} = 1,024$ Mbit/s

CMAN NO AI	Deliverable:
	`codefest/cf08/cman_aer_analysis.md` (or .pdf/.png) containing 1. mean aggregate spike rate R; 2. mean AER bandwidth B in Mbit/s with formula and substituted values; 3. interface comparison table (SPI, I²C, AXI4-Lite) with sustain Y/N for each and the chosen lowest-complexity interface; 4. burst-peak bandwidth, burst-to-mean ratio, and buffering decision; 5. frame-based bandwidth, AER/frame ratio at f=50 Hz, and crossover firing rate f_crossover with one-sentence implication.

3.)

Interface	Max bandwidth	Headroom = max/1.024	Sustainable?	Note
I^2C	≤ 3.4 Mbit/s	$3.4/1024 = 3.32x$	Yes	2-wire bus, simplest
SPI	≤ 50 Mbit/s	$50/1.024 = 49x$	Yes	4-wire, hard
AXI4-lite	100 Mbit/s	$100/1.024 = 98x$	Yes	Most complex

~~I²C is the least complex~~ SPI is better, can handle burst

4.) a $0.25 \times 1024 = 256$ Neurons, 256 spikes/1ms

4b.) $256 \times 20 = 5120$ bits

$B_{peak} = \frac{5120}{1ms} = 5120000$ bit/s

$Mbit/s = B_{peak} = 5.12$ Mbit/s

4c) $= \frac{B_{peak}}{B_{mean}} = \frac{5.12}{1.024} = 5.0x$

4d.) $\frac{50 Mbit/s}{5.12 Mbit/s} = 9.8x$ head room

4e.) $B_p \times 1ms = 5.12 \times 10^6 \times 10^{-3} = 5120$ bits

$B\_I^2C \times 1ms = 3.4 \times 10^6 \times 10^{-3} = 3,400$ bits

backlog $= 5120 - 3400 = 1720 = 86$ packets

5.) 5a $\frac{1}{T\_frame} = \frac{1}{0.001} = 1000$ frames/s

bits/frame $= N = 1024$

$B_{frame} = 1024 \times 1000 \times 1 = 1.024$ Mbit/s

5b.) $\frac{B_{AER}}{B_{frame}} = \frac{1.024}{1.024} = 1$

5c $= B_{AER}(f) = N \times f \times 20$

$B_{frame} = N\left(\frac{1}{T_{frame}}\right) \times 1 = N \times 1000$

$f = 50$ Hz